

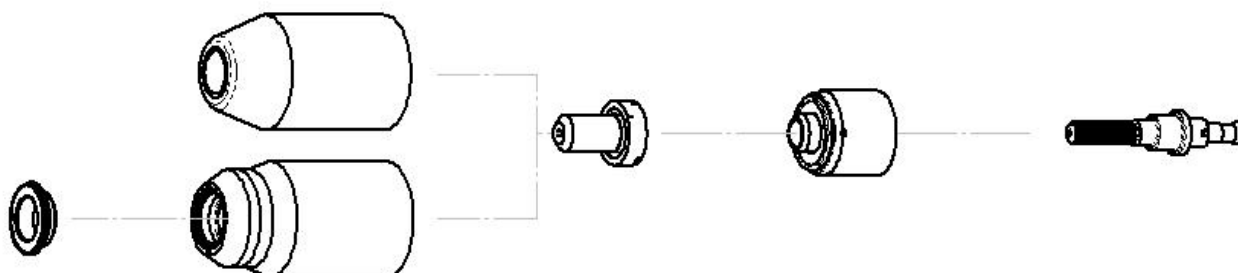
CUTMASTER A120

4T.08 Recommended Cutting Speeds for Machine and Automated Torches With Exposed Tip

Mild Steel

40A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

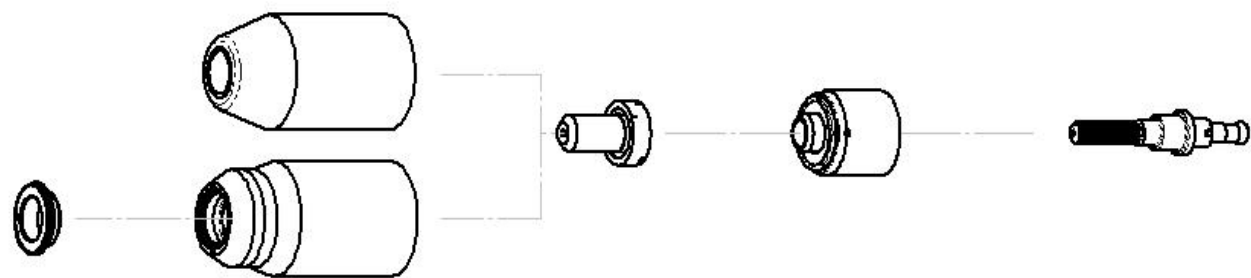
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.036	70 (25') 75 (50')	101	0.14	160	0.18	0.0	0.05
16		0.060		103	0.14	140	0.18	0.0	0.05
14		0.075		105	0.14	120	0.18	0.1	0.06
12		0.105		108	0.14	80	0.18	0.2	0.06
10		0.135		110	0.14	60	0.18	0.3	0.06
	3/16	0.188		111	0.14	55	0.18	0.4	0.06
	1/4	0.250		117	0.14	40	0.18	0.5	0.07
	3/8	0.375		119	0.14	25	0.18	1.2	0.07
	1/2	0.500		120	0.14	15	0.2	2.0	0.07

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	4.8 (7.6m) 5.2 (15.2m)	101	3.6	3990	4.6	0.0	1.1
2		105	3.6	2920	4.6	0.1	1.4
3		109	3.6	1810	4.6	0.3	1.5
4		110	3.6	1470	4.6	0.3	1.6
5		112	3.6	1345	4.6	0.4	1.6
6		116	3.6	1100	4.6	0.5	1.7
8		118	3.6	815	4.6	1.0	1.7
10		119	3.6	595	4.6	1.5	1.8
12		120	3.6	435	5.1	2.0	1.8

BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A120

Stainless Steel 40A Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

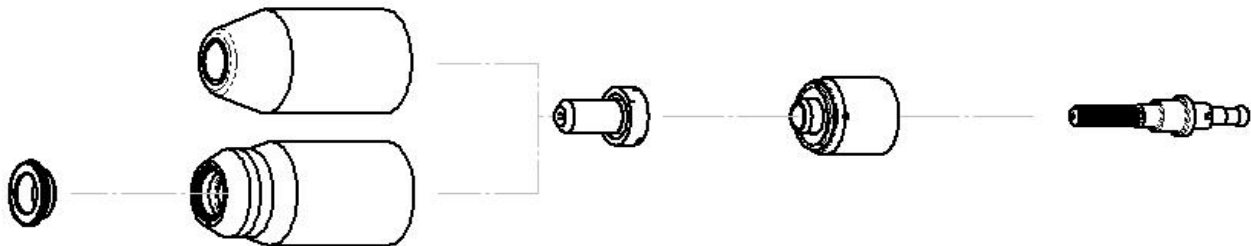
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
18		0.050	75 (25') 80 (50')	110	0.19	60	0.20	0.00	0.07
16		0.063		100	0.19	50	0.20	0.00	0.07
14		0.078		105	0.19	45	0.20	0.10	0.07
12		0.109		110	0.19	40	0.20	0.20	0.07
10		0.141		108	0.19	35	0.20	0.30	0.07
	3/16	0.188		110	0.19	30	0.20	0.40	0.07
	1/4	0.250		120	0.19	18	0.20	0.50	0.08
	3/8	0.375		126	0.19	10	0.20	2.00	0.08
	1/2	0.500		118	0.19	8	Edge start		0.09

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.2 (7.6) 5.5 (15.2)	112	4.8	1670	4.8	0.0	1.7
2		105	4.8	1140	5.1	0.1	1.8
3		109	4.8	980	5.1	0.2	1.8
4		109	4.8	845	5.1	0.3	1.8
5		111	4.8	715	5.1	0.4	1.8
6		118	4.8	525	5.1	0.5	2.0
8		123	4.8	350	5.1	1.5	2.0
10		125	4.8	245	5.1	2.0	2.0
12		120	4.8	215	<i>Edge Start</i>		2.2

BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Aluminum 40A Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.040	70 (25') 75 (50')	95	0.16	300	0.16	0.0	0.06
16		0.063		97	0.16	170	0.16	0.0	0.06
12		0.097		113	0.16	100	0.16	0.2	0.07
11		0.125		115	0.16	90	0.18	0.3	0.07
9		0.160		113	0.18	85	0.18	0.4	0.07
	3/16	0.188		116	0.18	75	0.18	0.5	0.07
	1/4	0.250		128	0.18	30	0.18	1.0	0.08
	3/8	0.375		150	0.18	10	Edge Start		0.09

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1.0	4.8 (7.6m) 5.2 (15.2m)	95	4.1	7620	4.1	0.0	1.5
2.0		104	4.1	3500	4.1	0.2	1.6
3.0		115	4.1	2350	4.6	0.3	1.7
4.0		113	4.6	2170	4.6	0.4	1.7
5.0		118	4.6	1740	4.6	0.5	1.8
6.0		125	4.6	1015	4.6	0.8	1.9
8.0		139	4.6	500	<i>Edge Start</i>		2.0
10.0		153	4.6	180	<i>Edge Start</i>		2.2

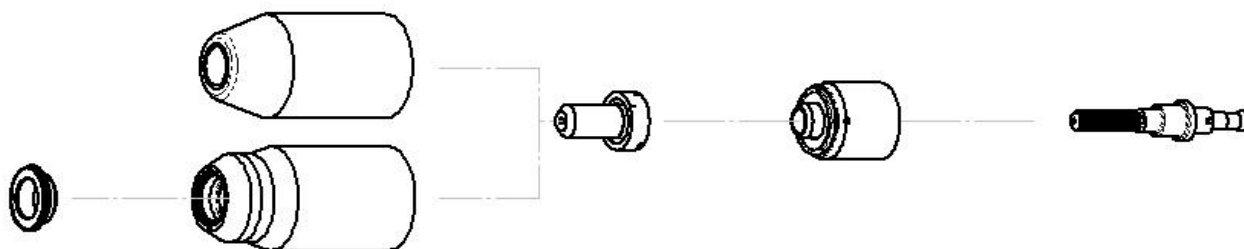
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Mild Steel

60A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25') 90 (50')	118	0.19	290	0.19	0.00	0.08
14		0.075		120	0.19	285	0.19	0.10	0.08
11		0.120		118	0.19	180	0.19	0.10	0.08
10		0.135		119	0.19	176	0.19	0.10	0.07
	3/16	0.188		121	0.19	100	0.19	0.20	0.08
	1/4	0.250		122	0.19	80	0.19	0.30	0.08
	3/8	0.375		124	0.19	50	0.19	0.50	0.09
	1/2	0.500		132	0.19	26	0.19	0.75	0.10
	5/8	0.625		135	0.19	19	Edge Start		0.10
	3/4	0.750		136	0.19	14	Edge Start		0.08
	1	1.000		150	0.19	6	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	115	4.8	7540	4.8	0	2.1
2		120	4.8	7015	4.0	0.10	1.9
3		118	4.8	4570	4.8	0.10	0.1
4		120	4.8	3650	4.8	0.20	1.9
5		121	4.8	2465	4.8	0.20	2.1
6		122	4.8	2145	4.8	0.30	2.0
8		123	4.8	1635	4.8	0.40	2.2
10		125	4.8	1180	4.8	0.60	2.4
12		130	4.8	795	4.8	0.75	2.4
15		134	4.8	530	<i>Edge Start</i>		2.4
20		138	4.8	325	<i>Edge Start</i>		2.2
25		149	4.8	165	<i>Edge Start</i>		2.7

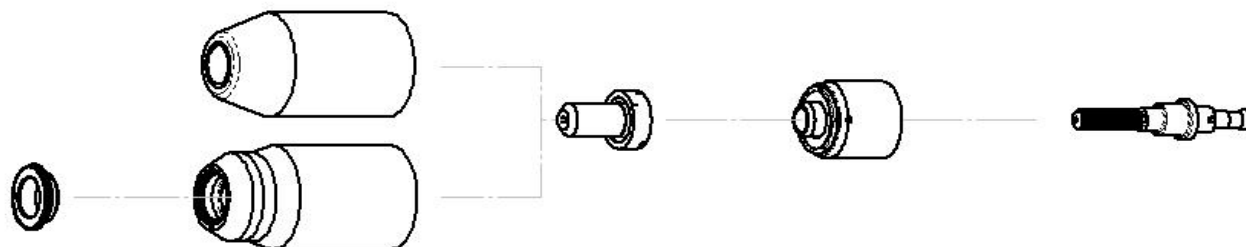
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Stainless Steel

60A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	119	0.19	350	0.20	0.00	0.05
14		0.078		116	0.19	300	0.20	0.10	0.07
11		0.125		123	0.19	150	0.20	0.10	0.07
10		0.141		118	0.19	125	0.20	0.10	0.08
	3/16	0.188		122	0.19	90	0.20	0.20	0.08
	1/4	0.250		123	0.19	65	0.20	0.30	0.09
	3/8	0.375		130	0.19	30	0.20	0.50	0.09
	1/2	0.500		132	0.19	21	0.20	0.90	0.08
	5/8	0.625		132	0.19	14	Edge Start		0.11
	3/4	0.750		135	0.19	10	Edge Start		0.10

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	124	4.8	10890	5.1	0.00	0.8
2		116	4.8	7560	5.1	0.10	1.7
3		122	4.8	4365	5.1	0.10	1.7
4		119	4.8	2865	5.1	0.20	2.1
5		122	4.8	2195	5.1	0.20	2.1
6		123	4.8	1790	5.1	0.30	2.2
8		127	4.8	1190	5.1	0.40	2.2
10		130	4.8	725	5.1	0.50	2.2
12		132	4.8	580	5.1	0.90	2.1
15		132	4.8	405	<i>Edge Start</i>		2.6
20		136	4.8	230	<i>Edge Start</i>		2.5

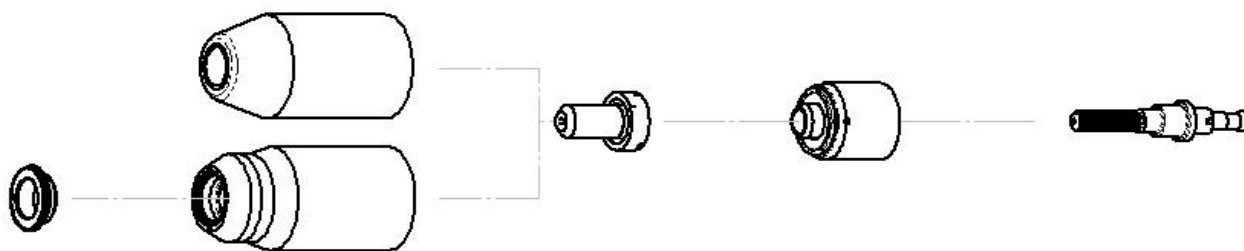
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Aluminum

60A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

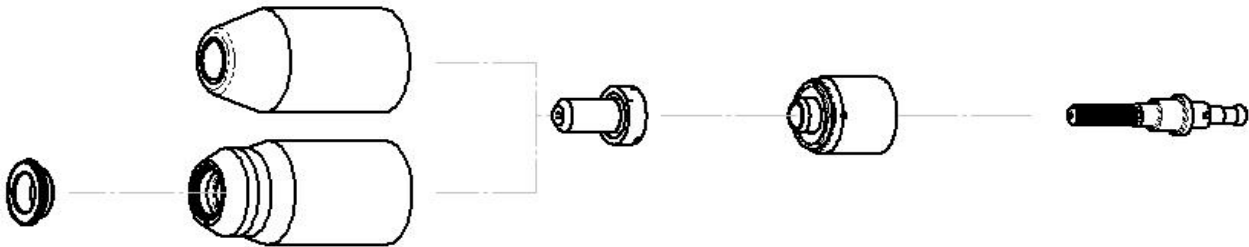
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	123	0.25	440	0.25	0.00	0.08
14		0.079		126	0.25	300	0.25	0.10	0.09
11		0.120		128	0.25	250	0.25	0.10	0.09
	3/16	0.188		132	0.25	170	0.25	0.20	0.09
	1/4	0.250		132	0.25	85	0.25	0.30	0.09
	3/8	0.375		141	0.25	45	0.25	0.50	0.10
	1/2	0.500		148	0.25	30	0.25	0.80	0.09
	5/8	0.625		145	0.19	18	Edge Start		0.08
	3/4	0.750		147	0.19	12	Edge Start		0.10

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	118	6.4	17010	6.4	0.00	1.8
2		126	6.4	7680	6.4	0.10	2.3
3		128	6.4	6410	6.4	0.10	2.3
4		130	6.4	5230	6.4	0.20	2.3
5		132	6.4	4010	6.4	0.20	2.4
6		132	6.4	2640	6.4	0.30	2.4
8		137	6.4	1630	6.4	0.40	2.4
10		142	6.4	1085	6.4	0.60	2.4
12		146	6.4	845	6.4	0.70	2.3
15		146	4.8	540	<i>Edge Start</i>		2.1
20		148	4.8	260	<i>Edge Start</i>		2.5

BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Mild Steel
80A
Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8211	9-8213	9-8232

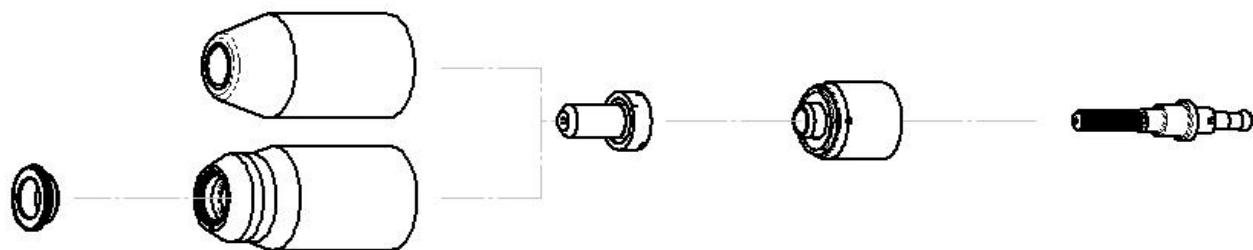
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25") 90 (50")	110	0.19	320	0.20	0.00	0.06
11		0.120		113	0.19	230	0.20	0.10	0.07
10		0.135		110	0.19	180	0.20	0.20	0.07
	3/16	0.188		110	0.19	136	0.20	0.30	0.07
	1/4	0.250		115	0.19	100	0.20	0.40	0.09
	3/8	0.375		125	0.19	42	0.20	0.50	0.09
	1/2	0.500		123	0.19	40	0.20	0.60	0.09
	5/8	0.625		133	0.19	18	0.20	0.75	0.10
	3/4	0.750		140	0.25	18	Edge Start		0.11
	7/8	0.875		150	0.25	10	Edge Start		0.11
	1	1.000		152	0.25	8	Edge Start		0.13

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	109	4.8	8915	5.1	0.00	1.5
2		111	4.8	7415	5.1	0.10	1.7
3		113	4.8	5915	5.1	0.10	1.8
4		110	4.8	4095	5.1	0.30	1.7
5		111	4.8	3325	5.1	0.30	1.8
6		114	4.8	2745	5.1	0.40	2.2
8		120	4.8	1775	5.1	0.50	2.3
10		125	4.8	1060	5.1	0.50	2.3
12		123	4.8	1025	5.1	0.60	2.3
15		130	4.8	610	5.1	0.75	2.5
20		143	6.4	395	Edge Start		2.7
25		152	6.4	210	Edge Start		3.2

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Stainless Steel 80A Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8211	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	110	0.19	340	0.25	0.00	0.06
11		0.125		115	.0.19	300	0.25	0.10	0.06
10		0.141		115	0.19	280	0.25	0.10	0.06
	3/16	0.188		115	0.19	140	0.25	0.20	0.07
	1/4	0.250		118	0.19	100	0.25	0.30	0.08
	3/8	0.375		119	0.19	45	0.25	0.40	0.08
	1/2	0.500		124	0.19	26	0.25	0.80	0.10
	5/8	0.625		133	0.19	16	Edge Start		0.10
	3/4	0.750		136	0.19	10	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	108	4.8	9020	6.4	0.00	1.5
2		111	4.8	8380	6.4	0.00	1.5
3		114	4.8	7730	6.4	0.10	1.5
4		115	4.8	5865	6.4	0.20	1.6
5		115	4.8	3410	6.4	0.20	1.8
6		117	4.8	2765	6.4	0.30	1.9
8		119	4.8	1815	6.4	0.40	2.0
10		120	4.8	1070	6.4	0.60	2.1
12		123	4.8	765	6.4	0.80	2.3
15		131	4.8	475	Edge Start		2.5
20		137	4.8	205	Edge Start		3.0

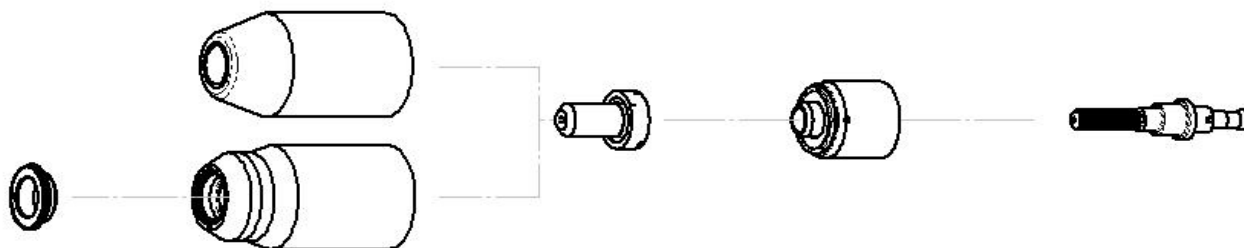
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Aluminum

80A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8211	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	116	0.25	350	0.25	0.00	0.10
11		0.120		120	0.25	280	0.25	0.10	0.10
	3/16	0.188		124	0.25	180	0.25	0.20	0.10
	1/4	0.250		130	0.25	110	0.25	0.30	0.09
	3/8	0.375		136	0.25	55	0.25	0.40	0.11
	1/2	0.500		139	0.25	38	0.25	0.60	0.11
	5/8	0.625		136	0.19	26	0.25	0.75	0.10
	3/4	0.750		150	0.19	14	Edge Start		0.12
	7/8	0.875		153	0.19	10	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	114	6.4	8890	6.4	0.00	2.4
2		117	6.4	8420	6.4	0.00	2.5
3		120	6.4	7170	6.4	0.10	2.5
4		122	6.4	5710	6.4	0.20	2.5
5		125	6.4	4315	6.4	0.20	2.5
6		129	6.4	3190	6.4	0.30	2.4
8		133	6.4	2070	6.4	0.40	2.5
10		136	6.4	1330	6.4	0.50	2.7
12		138	6.4	1060	6.4	0.50	2.9
15		137	4.8	745	6.4	0.75	2.5
20		151	4.8	325	Edge Start		3.0

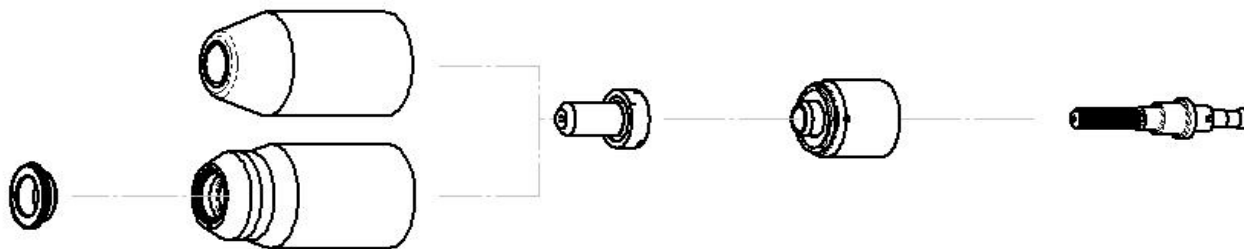
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Mild Steel

100A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8212	9-8213	9-8232

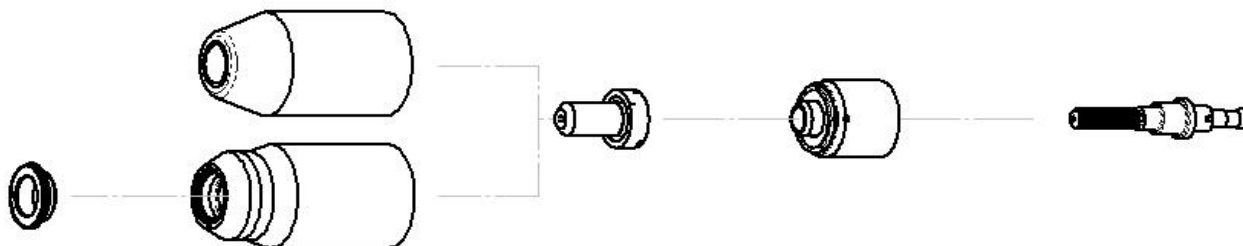
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	110	0.18	85	0.20	0.10	0.09
	3/8	0.375		112	0.18	75	0.20	0.20	0.09
	1/2	0.500		115	0.18	45	0.20	0.40	0.10
	5/8	0.625		118	0.18	30	0.20	0.60	0.11
	3/4	0.750		120	0.18	20	0.20	1.20	0.12
	1	1.000		123	0.18	15	Edge Start		0.12
	1-1/4	1.250		128	0.18	8	Edge Start		0.13
Material Thickness				Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6			5.2 (7.6m) 5.2 (15.2m)	110	4.6	2200	5.1	0.10	2.3
8				111	4.6	2030	5.1	0.10	2.3
10				112	4.6	1790	5.1	0.20	2.3
12				114	4.6	1310	5.1	0.40	2.5
15				117	4.6	870	5.1	0.60	2.7
20				120	4.6	490	5.0	1.20	3.0
25				123	4.6	390	Edge Start		3.0
30				127	4.6	250	Edge Start		3.2

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

Stainless Steel

100A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8212	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	102	0.16	100	0.18	0.00	0.09
	3/8	0.375		106	0.16	65	0.18	0.10	0.10
	1/2	0.500		109	0.16	45	0.18	0.30	0.10
	5/8	0.625		123	0.19	20	0.23	2.00	0.11
	3/4	0.750		128	0.19	15	Edge Start		0.12
	1	1.000		135	0.19	10	Edge Start		0.13
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6			5.2 (7.6m) 5.2 (15.2m)	102	4.1	2630	4.6	0.00	2.3
8				104	4.1	2080	4.6	0.10	2.4
10				106	4.1	1575	4.6	0.20	2.4
12				108	4.1	1255	4.6	0.30	2.4
15				119	4.6	685	5.7	2.00	2.7
20				129	4.8	360	Edge Start		3.1
25				135	4.8	260	Edge Start		3.2

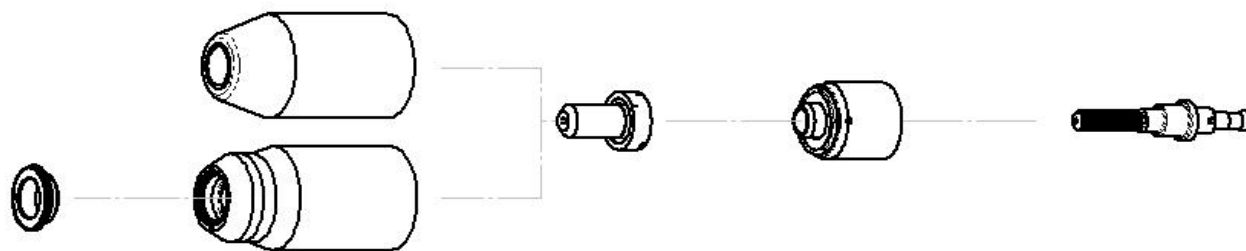
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Aluminum

100A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8212	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	110	0.19	100	0.20	0.00	0.09
	3/8	0.375		124	0.19	65	0.20	0.20	0.10
	1/2	0.500		128	0.19	45	0.20	0.30	0.11
	5/8	0.625		132	0.19	35	0.20	0.50	0.11
	3/4	0.750		135	0.19	20	0.20	2.00	0.13
	1	1.000		140	0.19	10	Edge Start		0.13
Material Thickness				Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6			5.2 (7.6m) 5.2 (15.2m)	109	4.8	2630	5.1	0.00	2.3
8				117	4.8	2080	5.1	0.10	2.5
10				125	4.8	1575	5.1	0.20	2.7
12				127	4.8	1255	5.1	0.30	2.7
15				131	4.8	960	5.1	0.50	2.8
20				136	4.8	470	6.0	2.00	3.3
25				140	4.8	270	Edge Start		3.4

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

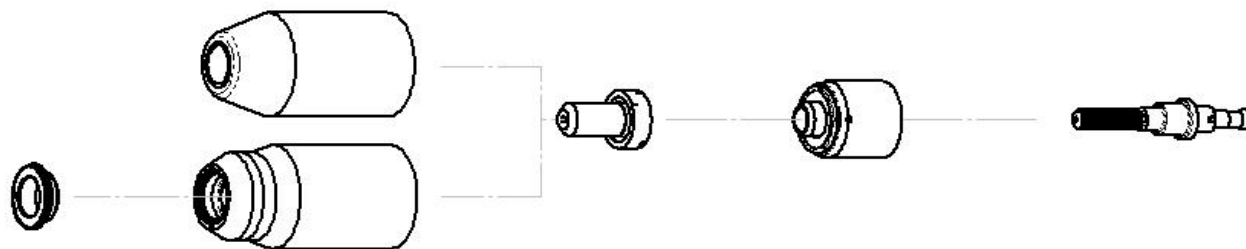
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CUTMASTER A120

Mild Steel

120A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8233	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	134	0.19	150	0.25	0.10	0.11
	3/8	0.375		141	0.19	85	0.25	0.30	0.11
	1/2	0.500		141	0.19	70	0.25	0.80	0.12
	5/8	0.625		146	0.19	45	0.25	1.00	0.12
	3/4	0.750		150	0.19	30	0.25	1.20	0.12
	7/8	0.875		160	0.25	25	Edge Start		0.14
	1	1.000		163	0.25	19	Edge Start		0.14
	1-1/4	1.250		168	0.25	12	Edge Start		0.15
	1-1/2	1.500		172	0.25	8	Edge Start		0.16

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6.0	5.5 (7.6m) 5.5 (15.2m)	133	4.8	3990	6.4	0.1	2.5
8.0		138	4.8	2950	6.4	0.2	2.7
10.0		141	4.8	2100	6.4	0.4	2.8
12.0		141	4.8	1860	6.4	0.6	2.9
15.0		145	4.8	1320	6.4	1.0	3.0
20.0		153	4.8	720	6.4	1.2	3.2
25.0		163	6.4	500	<i>Edge Start</i>		3.6
30.0		167	6.4	355	<i>Edge Start</i>		3.8
35.0		170	6.4	255	<i>Edge Start</i>		4.0
40.0		173	6.4	175	<i>Edge Start</i>		4.2

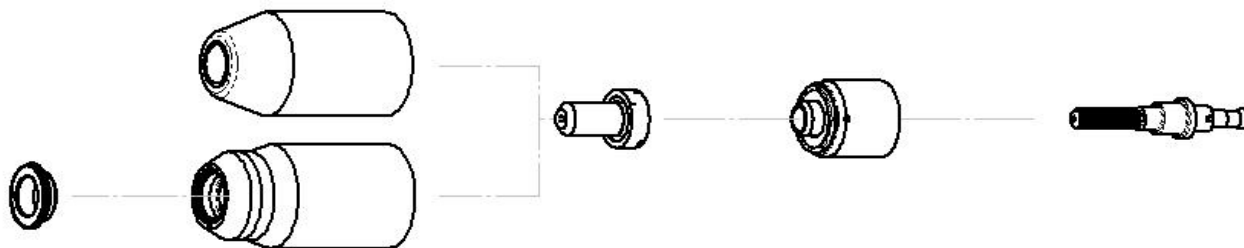
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Stainless Steel

120A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8233	9-8213	9-8232

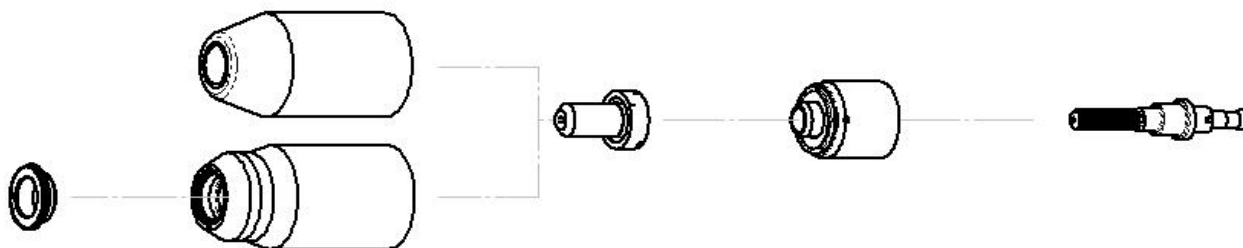
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	133	0.19	180	0.25	0.20	0.10
	3/8	0.375		136	0.19	100	0.25	0.40	0.11
	1/2	0.500		140	0.19	60	0.25	0.80	0.11
	5/8	0.625		152	0.25	40	0.30	1.20	0.13
	3/4	0.750		154	0.25	26	Edge Start		0.14
	1	1.000		159	0.25	16	Edge Start		0.14
	1-1/4	1.250		167	0.25	8	Edge Start		0.14
Material Thickness				Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6			5.5 (7.6m) 5.5 (15.2m)	133	4.8	4800	6.4	0.2	2.6
8				135	4.8	3520	6.4	0.2	2.7
10				137	4.8	2390	6.4	0.5	2.8
12				139	4.8	1750	6.4	0.8	2.9
15				149	6.4	1160	7.6	1.2	3.4
20				155	6.4	625	Edge Start		3.6
25				159	6.4	425	Edge Start		3.6
30				165	6.4	260	Edge Start		3.6

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

Aluminum

120A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8233	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	139	0.19	180	0.25	0.30	0.11
	3/8	0.375		140	0.19	110	0.25	0.50	0.11
	1/2	0.500		143	0.19	75	0.25	0.80	0.11
	5/8	0.625		156	0.25	50	0.30	1.00	0.12
	3/4	0.750		161	0.25	34	Edge Start		0.13
	1	1.000		167	0.25	23	Edge Start		0.14
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6			5.5 (7.6m) 5.5 (15.2m)	139	4.8	4770	6.4	0.2	2.8
8				140	4.8	3650	6.4	0.2	2.8
10				140	4.8	2660	6.4	0.5	2.8
12				142	4.8	2100	6.4	0.8	2.8
15				152	6.4	1445	7.6	1.2	3.0
20				162	6.4	820	Edge Start		3.4
25				167	6.4	600	Edge Start		3.4

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

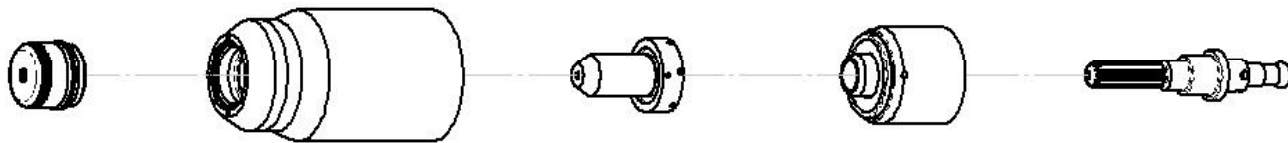
CUTMASTER A120

4T.09 Recommended Cutting Speeds for Machine and Automated Torches With Shielded Tip

Mild Steel

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

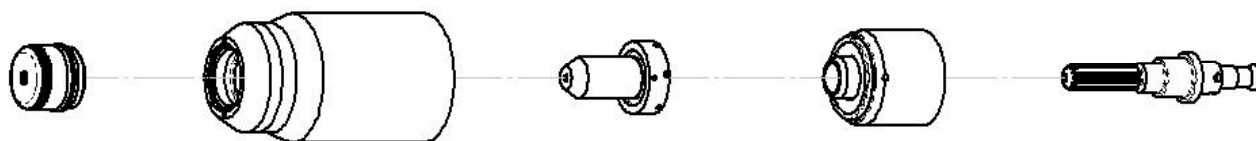
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.036	75 (25') 80 (50')	107	0.16	130	0.16	0.0	0.06
16		0.060		113	0.16	120	0.16	0.0	0.06
14		0.075		108	0.16	90	0.16	0.1	0.06
12		0.105		111	0.16	75	0.16	0.2	0.06
10		0.135		114	0.16	65	0.16	0.3	0.07
	3/16	0.188		115	0.16	65	0.16	0.4	0.07
	1/4	0.250		118	0.16	45	0.16	0.5	0.07
	3/8	0.375		123	0.16	23	0.16	1.2	0.08
	1/2	0.500		128	0.16	18	0.16	2.0	0.08
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1			5.2 (7.6) 5.5 (15.2)	108	4.1	3266	4.1	0.0	1.4
2				108	4.1	2239	4.1	0.0	1.5
3				112	4.1	1794	4.1	0.1	1.7
4				114	4.1	1651	4.1	0.2	1.7
5				115	4.1	1578	4.1	0.3	1.7
6				117	4.1	1256	4.1	0.4	1.7
8				121	4.1	853	4.1	0.5	1.7
10				124	4.1	565	4.1	1.2	1.8
12				127	4.1	485	4.1	2.0	1.9

BOLD TYPE indicates maximum piercing parameters.

Stainless Steel

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
18		0.050	75 (25') 80 (50')	110	0.16	60	4.1	0.0	0.06
16		0.063		108	0.16	50	4.1	0.1	0.07
14		0.078		114	0.16	45	4.1	0.1	0.07
12		0.109		113	0.16	40	4.1	0.2	0.07
10		0.141		116	0.16	35	4.1	0.3	0.07
	3/16	0.188		115	0.16	30	4.1	0.4	0.07
	1/4	0.250		118	0.16	20	4.1	1.2	0.07
	3/8	0.375		125	0.16	15	4.1	1.8	0.08
	1/2	0.500		127	0.16	10	4.1	2.0	0.08
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1			5.2 (7.6) 5.5 (15.2)	109	4.1	1670	4.1	0.0	1.7
2				114	4.1	1140	4.1	0.1	1.8
3				114	4.1	980	4.1	0.2	1.8
4				116	4.1	845	4.1	0.3	1.8
5				115	4.1	725	4.1	0.4	1.8
6				117	4.1	565	4.1	0.5	2.0
8				122	4.1	440	4.1	1.5	2.0
10				125	4.1	360	4.1	1.8	2.0
12				127	4.1	280	4.1	2.0	2.2

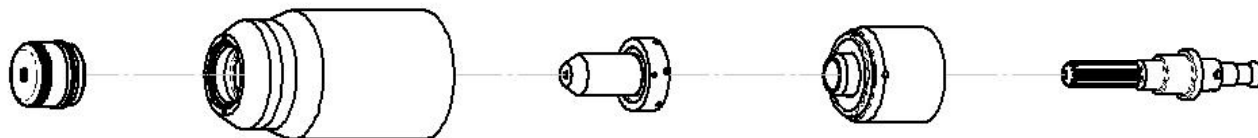
BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A120

Aluminum

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed	
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)	
20		0.040	75 (25') 80 (50')	110	0.19	300	0.20	0.0	0.06	
16		0.063		113	0.19	170	0.20	0.1	0.07	
12		0.097		120	0.19	100	0.20	0.2	0.07	
11		0.125		125	0.19	90	0.20	0.3	0.07	
9		0.160		126	0.19	85	0.20	0.4	0.07	
	3/16	0.188		128	0.19	70	0.20	0.5	0.07	
	1/4	0.250		137	0.19	30	0.20	1.0	0.08	
	3/8	0.375		145	0.19	10	0.20	2.0	0.09	
Material Thickness				Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)				Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1.0			5.2 (7.6) 5.5 (15.2)	110	4.8	7660	5.1	0.0	1.6	
2.0				116	4.8	3490	5.1	0.2	1.8	
3.0				124	4.8	2350	5.1	0.3	1.8	
4.0				126	4.8	2170	5.1	0.4	1.8	
5.0				129	4.8	1630	5.1	0.6	1.8	
6.0				135	4.8	990	5.1	1.0	1.9	
8.0				141	4.8	500	5.1	1.6	2.0	
10.0				146	4.8	180	5.1	2.1	2.3	

BOLD TYPE indicates maximum piercing parameters.

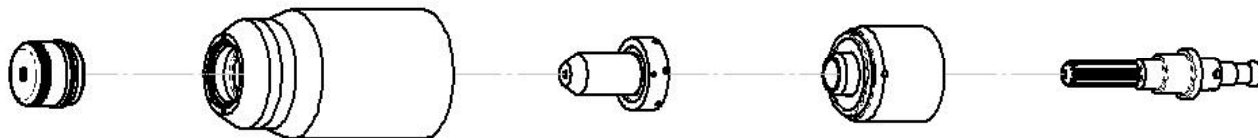
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CUTMASTER A120

Mild Steel

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25') 90 (50')	128	0.19	250	0.20	0.00	0.09
14		0.075		129	0.19	237	0.20	0.10	0.09
11		0.120		126	0.19	200	0.20	0.10	0.08
10		0.135		128	0.19	142	0.20	0.10	0.09
	3/16	0.188		132	0.19	115	0.20	0.20	0.09
	1/4	0.250		134	0.19	80	0.20	0.30	0.08
	3/8	0.375		132	0.19	34	0.20	0.70	0.10
	1/2	0.500		144	0.19	23	0.20	0.75	0.10
	5/8	0.625		150	0.19	14	Edge Start		0.10
	3/4	0.750		145	0.19	14	Edge Start		0.10
	1	1.000		160	0.19	4	Edge Start		0.12

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	127	4.8	6804	5.1	0	2.2
2		129	4.8	5942	5.1	0.10	2.2
3		126	4.8	5080	5.1	0.10	0.1
4		130	4.8	3316	5.1	0.20	2.3
5		132	4.8	2794	5.1	0.20	2.2
6		134	4.8	2230	5.1	0.30	2.1
8		133	4.8	1425	5.1	0.40	2.3
10		134	4.8	822	5.1	0.60	2.5
12		141	4.8	646	5.1	0.75	2.5
15		148	4.8	419	<i>Edge Start</i>		2.6
20		147	4.8	318	<i>Edge Start</i>		2.7
25		159	4.8	118	<i>Edge Start</i>		3.0

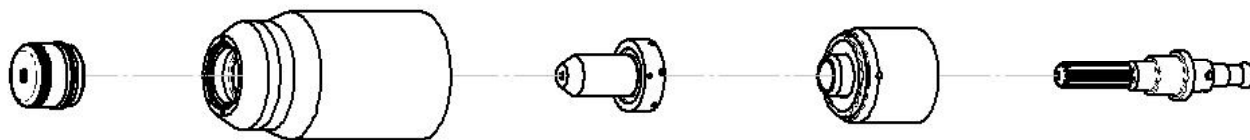
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Stainless Steel

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	110	0.13	165	0.20	0.00	0.06
14		0.078		116	0.13	155	0.20	0.10	0.07
11		0.125		118	0.13	125	0.20	0.10	0.07
10		0.141		126	0.13	80	0.20	0.10	0.09
	3/16	0.188		125	0.13	75	0.20	0.20	0.09
	1/4	0.250		127	0.13	60	0.20	0.30	0.08
	3/8	0.375		134	0.13	28	0.20	0.50	0.08
	1/2	0.500		136	0.19	17	0.25	0.75	0.09
	5/8	0.625		131	0.13	14	Edge Start		0.08
	3/4	0.750		142	0.19	10	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	101	3.3	4590	5.1	0.00	1.2
2		116	3.3	3925	5.1	0.10	1.7
3		118	3.3	3285	5.1	0.10	1.7
4		126	3.3	1985	5.1	0.20	2.2
5		125	3.3	1850	5.1	0.20	2.1
6		127	3.3	1605	5.1	0.30	2.1
8		131	3.3	1100	5.1	0.40	2.1
10		134	3.3	670	5.1	0.50	2.1
12		136	4.8	490	6.4	0.90	2.2
15		132	3.3	375	<i>Edge Start</i>		2.6
20		144	4.8	230	<i>Edge Start</i>		2.5

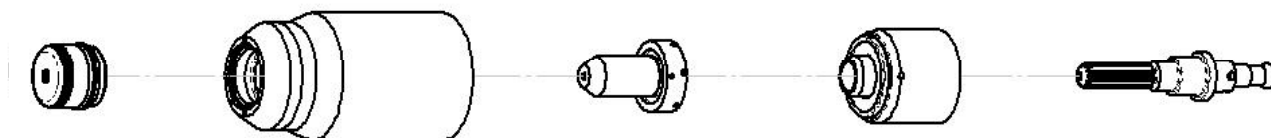
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Aluminum

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	118	0.13	350	0.20	0.00	0.07
14		0.079		118	0.13	350	0.20	0.10	0.08
11		0.120		123	0.13	275	0.20	0.10	0.08
	3/16	0.188		125	0.13	140	0.20	0.20	0.08
	1/4	0.250		132	0.13	80	0.20	0.30	0.08
	3/8	0.375		135	0.13	45	0.20	0.50	0.08
	1/2	0.500		140	0.13	26	0.20	0.80	0.08
	5/8	0.625		148	0.19	18	Edge start		0.10
	3/4	0.750		155	0.19	12	Edge start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	118	6.4	8890	6.4	0.00	2
2		118	6.4	8890	6.4	0.10	2.0
3		123	6.4	7070	6.4	0.10	2.1
4		124	6.4	5095	6.4	0.20	2.0
5		126	6.4	3335	6.4	0.20	2.0
6		130	6.4	2370	6.4	0.30	2.0
8		134	6.4	1570	6.4	0.40	2.0
10		136	6.4	1070	6.4	0.60	2.0
12		139	6.4	765	6.4	0.70	2.0
15		146	4.8	515	<i>Edge start</i>		2.4
20		157	4.8	260	<i>Edge start</i>		2.7

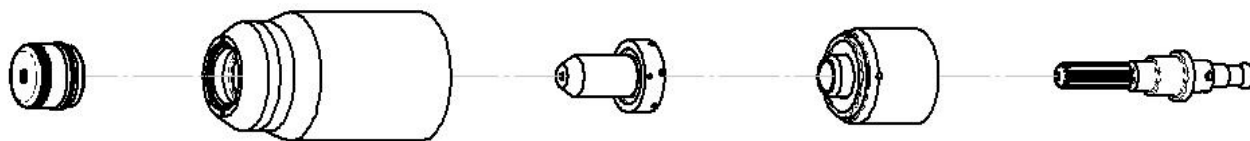
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Mild Steel

80A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8211	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25') 90 (50')	115	0.13	280	0.20	0.00	0.06
11		0.120		112	0.13	190	0.20	0.10	0.06
10		0.135		110	0.13	176	0.20	0.20	0.06
	3/16	0.188		122	0.19	135	0.20	0.30	0.08
	1/4	0.250		127	0.19	100	0.20	0.40	0.09
	3/8	0.375		134	0.19	40	0.20	0.50	0.10
	1/2	0.500		136	0.19	36	0.20	0.60	0.10
	5/8	0.625		145	0.19	18	Edge Start		0.11
	3/4	0.750		147	0.19	14	Edge Start		0.11
	7/8	0.875		146	0.19	10	Edge Start		0.12
	1.000	1.000		152	0.187	8	Edge Start		0.13

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	116	3.3	7895	5.1	0.00	1.6
2		114	3.3	6395	5.1	0.10	1.6
3		112	3.3	4895	5.1	0.10	1.5
4		115	4.8	4025	5.1	0.30	1.7
5		123	4.8	3300	5.1	0.30	2.0
6		126	4.8	2735	5.1	0.40	2.2
8		131	4.8	1745	5.1	0.50	2.5
10		134	4.8	1000	5.1	0.50	2.6
12		136	4.8	935	5.1	0.60	2.6
15		143	4.8	580	5.1	0.80	2.8
20		147	4.8	325	<i>Edge Start</i>		2.9
25		151	4.8	210	<i>Edge Start</i>		3.2

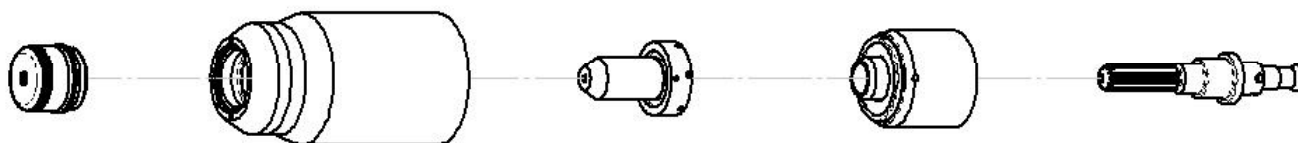
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Stainless Steel

80A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8211	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	114	0.125	340	0.2	0.00	0.06
11		0.125		115	0.125	260	0.2	0.10	0.06
10		0.141		116	0.125	250	0.2	0.10	0.06
	3/16	0.188		115	0.125	170	0.2	0.20	0.07
	1/4	0.250		118	0.125	85	0.2	0.30	0.08
	3/8	0.375		127	0.19	45	0.25	0.40	0.09
	1/2	0.500		136	0.19	16	0.25	0.75	0.11
	5/8	0.625		143	0.19	10	Edge Start		0.12
	3/4	0.750		146	0.19	8	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	114	3.2	9410	5.1	0.00	1.5
2		114	3.2	8120	5.1	0.00	1.4
3		115	3.2	6830	5.1	0.10	1.4
4		116	3.2	5635	5.1	0.20	1.5
5		115	3.2	4010	5.1	0.20	1.8
6		117	3.2	2640	5.1	0.30	2.0
8		123	4.8	1630	6.4	0.40	2.1
10		128	4.8	1030	6.4	0.60	2.2
12		134	4.8	565	6.4	0.75	2.6
15		141	4.8	295	<i>Edge Start</i>		3.0
20		147	4.8	185	<i>Edge Start</i>		2.8

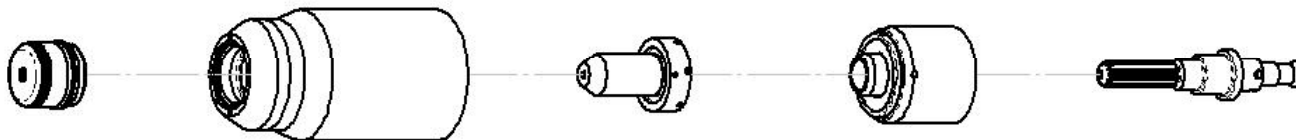
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Aluminum

80A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8211	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	115	0.13	320	0.2	0.00	0.09
11		0.120		120	0.13	240	0.2	0.10	0.08
	3/16	0.188		122	0.13	165	0.2	0.20	0.08
	1/4	0.250		124	0.13	100	0.2	0.30	0.10
	3/8	0.375		140	0.19	60	0.2	0.40	0.09
	1/2	0.500		142	0.19	36	0.2	0.60	0.11
	5/8	0.625		148	0.19	26	0.2	0.75	0.11
	3/4	0.750		155	0.19	16	Edge Start		0.12
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)

1	5.9 (7.6m) 6.2 (15.2m)	113	3.3	9020	5.1	0.00	2.4
2		116	3.3	7595	5.1	0.00	2.2
3		120	3.3	6165	5.1	0.10	2.0
4		121	3.3	5045	5.1	0.20	2.1
5		122	3.3	3955	5.1	0.20	2.2
6		124	3.3	2905	5.1	0.30	2.3
8		132	4.8	2010	5.1	0.40	2.3
10		140	4.8	1430	5.1	0.50	2.3
12		142	4.8	1045	5.1	0.60	2.6
15		146	4.8	730	5.1	0.75	2.7
20		157	4.8	330	<i>Edge Start</i>		3.2

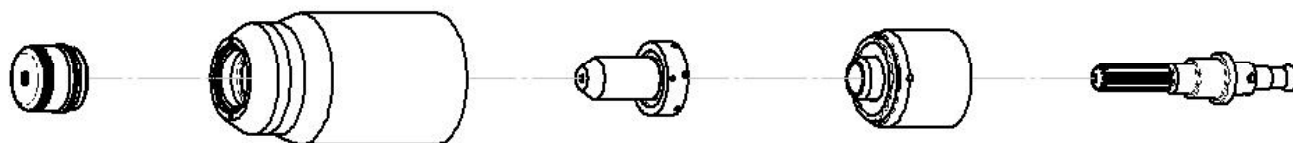
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A120

Mild Steel

100A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8233	9-8213	9-8232

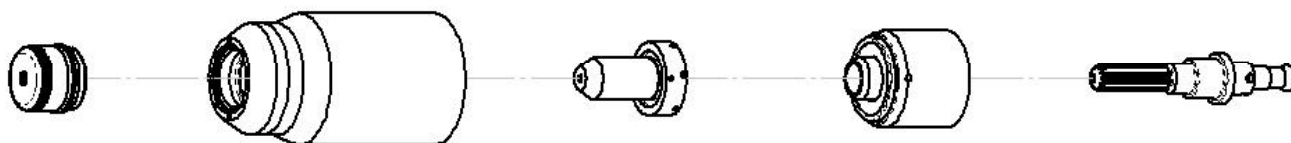
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	114	0.16	85	0.18	0.00	0.08
	3/8	0.375		112	0.16	70	0.18	0.20	0.09
	1/2	0.500		115	0.16	40	0.18	0.30	0.10
	5/8	0.625		123	0.16	30	0.18	0.40	0.10
	3/4	0.750		127	0.16	20	0.18	0.80	0.11
	1	1.000		133	0.16	12	Edge Start		0.12
	1-1/4	1.250		135	0.16	8	Edge Start		0.12
Material Thickness				Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay
(mm)			Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6				114	4.1	2210	4.6	0.00	2.0
8				113	4.1	1960	4.6	0.10	2.2
10				112	4.1	1665	4.6	0.20	2.3
12				114	4.1	1185	4.6	0.30	2.4
15				121	4.1	830	4.6	0.40	2.5
20				128	4.1	475	5.0	0.80	2.8
25				133	4.1	315	Edge Start		2.9
30				134	4.1	230	Edge Start		3.1

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

Stainless Steel

100A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8212	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	114	0.16	110	0.18	0.00	0.08
	3/8	0.375		116	0.16	70	0.18	0.10	0.09
	1/2	0.500		119	0.16	45	0.18	0.30	0.10
	5/8	0.625		130	0.19	20	0.23	2.00	0.11
	3/4	0.750		135	0.19	15	0.23	2.80	0.11
	1	1.000		140	0.19	10	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6	5.2 (7.6m) 5.2 (15.2m)	114	4.1	2900	4.6	0.00	2.0
8		115	4.1	2265	4.6	0.10	2.2
10		116	4.1	1685	4.6	0.20	2.4
12		118	4.1	1285	4.6	0.30	2.4
15		127	4.6	685	5.7	2.00	2.7
20		136	4.8	360	5.7	2.80	2.8
25		140	4.8	260	Edge Start		2.8

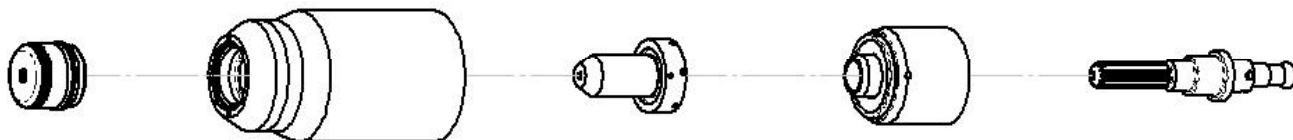
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Aluminum

100A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8239	9-8237	9-8212	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	75 (25') 75 (50')	121	0.18	110	0.19	0.00	0.10
	3/8	0.375		129	0.18	60	0.19	0.20	0.10
	1/2	0.500		135	0.18	45	0.19	0.30	0.11
	5/8	0.625		133	0.18	32	0.19	0.60	0.11
	3/4	0.750		138	0.18	25	0.19	2.00	0.11
	1	1.000		143	0.18	15	Edge Start		0.14

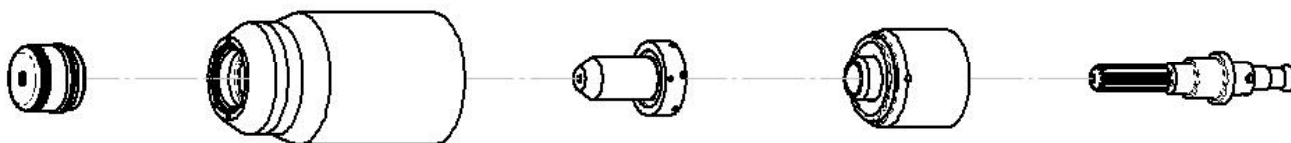
Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6	5.2 (7.6m) 5.2 (15.2m)	120	4.6	2920	4.8	0.00	2.5
8		125	4.6	2135	4.8	0.10	2.5
10		130	4.6	1470	4.8	0.20	2.6
12		134	4.6	1230	4.8	0.30	2.7
15		134	4.6	905	4.8	0.60	2.9
20		139	4.6	595	6.0	2.00	3.0
25		143	4.6	395	Edge Start		3.6

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

Mild Steel

120A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8256	9-8237	9-8233	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	139	0.125	160	0.20	0.20	0.11
	3/8	0.375		140	0.125	85	0.20	0.20	0.11
	1/2	0.500		142	0.125	75	0.20	0.50	0.10
	5/8	0.625		144	0.125	45	0.20	0.70	0.11
	3/4	0.750		150	0.125	30	0.25	1.50	0.12
	7/8	0.875		158	0.175	25	Edge Start		0.13
	1	1.000		160	0.175	22	Edge Start		0.14
	1-1/4	1.250		165	0.175	15	Edge Start		0.17
	1-1/2	1.500		171	0.175	8	Edge Start		0.15

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6	5.5 (7.6m) 5.5 (15.2m)	139	4.8	4270	5.08	0.20	2.8
8		140	4.8	3070	5.1	0.2	2.8
10		140	6.4	2120	5.08	0.30	2.8
12		142	6.4	1960	5.08	0.50	2.8
15		143	4.8	1355	5.08	0.70	2.8
20		152	4.8	725	6.4	1.6	3.1
25		160	4.8	570	Edge Start		3.6
30		164	4.8	430	Edge Start		4.0
35		168	4.8	290	Edge Start		4.3
40		173	4.45	150	Edge Start		4.4

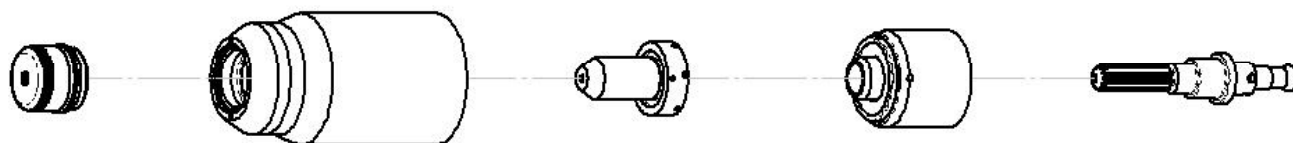
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A120

Stainless Steel

120A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8256	9-8237	9-8233	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	140	0.13	180	0.20	0.30	0.11
	3/8	0.375		140	0.13	100	0.20	0.40	0.12
	1/2	0.500		142	0.15	65	0.25	0.80	0.12
	5/8	0.625		152	0.15	40	0.25	1.00	0.12
	3/4	0.750		150	0.15	26	Edge Start		0.13
	1	1.000		155	0.15	16	Edge Start		0.13
	1-1/4	1.250		165	0.15	8	Edge Start		0.13

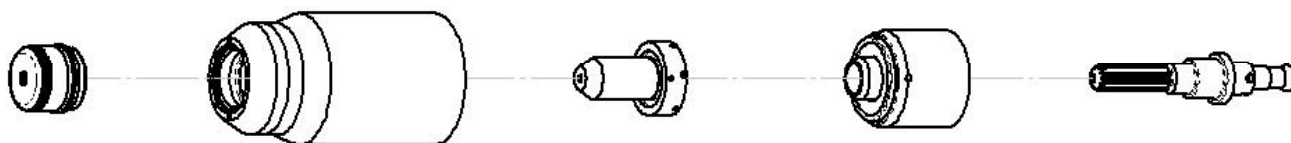
Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6	5.5 (7.6m) 5.5 (15.2m)	140	4.8	4800	5.1	0.3	2.8
8		140	4.8	3520	5.1	0.4	2.7
10		140	4.8	2410	5.1	0.6	3.0
12		142	4.8	1850	6.4	0.8	3.1
15		149	6.4	1190	6.4	1.0	3.1
20		151	6.4	620	Edge Start		3.3
25		155	6.4	420	Edge Start		3.3
30		162	6.4	260	Edge Start		3.3

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

Aluminum

120A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8256	9-8237	9-8233	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
	1/4	0.250	80 (25') 80 (50')	140	0.13	180	0.20	0.20	0.10
	3/8	0.375		142	0.13	110	0.20	0.30	0.11
	1/2	0.500		148	0.13	75	0.20	0.50	0.11
	5/8	0.625		156	0.15	45	0.25	0.80	0.11
	3/4	0.750		158	0.15	35	Edge Start		0.11
	1	1.000		164	0.15	23	Edge Start		0.12

Material Thickness		Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)		Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
6	5.5 (7.6m) 5.5 (15.2m)	5.5 (7.6m) 5.5 (15.2m)	140	4.8	4770	5.1	0.2	2.6
8			141	4.8	3650	5.1	0.3	2.6
10			143	4.8	2660	5.1	0.4	2.7
12			147	4.8	2100	5.1	0.5	2.8
15			154	4.8	1355	6.4	0.8	2.8
20			159	4.8	845	Edge Start		2.9
25			164	4.8	605	Edge Start		3.0

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

PATENT INFORMATION

Plasma Cutting Torch Patents

The following parts are covered under U.S. and Foreign Patents as follows:

Catalog #	Description	Patent(s)
9-8215	Electrode	US Pat No(s) 6163008; 6987238 Other Pat(s) Pending
9-8232	Electrode	US Pat No(s) 6163008; 6987238 Other Pat(s) Pending
9-8213	Cartridge	US Pat No(s) 6903301; 6717096; 6936786; 6703581; D496842; D511280; D492709; D499620; D504142 Other Pat(s) Pending
9-8205	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8206	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8207	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8252	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8208	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8209	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8210	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8231	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8211	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8212	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8253	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8233	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8225	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8226	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8227	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8228	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8241	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8243	Shield Cap	US Pat No(s) 6914211; D493183 Other Pat(s) Pending
9-8235	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8236	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8237	Shield Cup	US Pat No(s) 6914211; D501632; D511633 Other Pat(s) Pending
9-8238	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending

<u>Catalog #</u>	<u>Description</u>	<u>Patent(s)</u>
9-8239	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending
9-8244	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8245	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending

The following parts are also licensed under U.S. Patent No. 5,120,930 and 5,132,512:

<u>Catalog #</u>	<u>Description</u>
9-8235	Shield Cap
9-8236	Shield Cap
9-8237	Shield Cup
9-8238	Shield Cap
9-8239	Shield Cap
9-8244	Shield Cap
9-8245	Shield Cap